Summary



- The term correlation refers to the degree of relationship between two or more variables.
- Scatter diagram is a graphic device for finding correlation between two variables.
- Karl Pearson correlation coefficient $r(x,y)=\frac{\operatorname{cov}(X,Y)}{\sigma_x \sigma_y}$
- Correlation coefficient r lies between -1 and 1. (i.e) $-1 \le r \le 1$
- When $r=+1$, then the correlation is perfect positive
- When $r=-1$, then the correlation is perfect negative
- When $r=0$, then there is no relationship between the variables, (i.e) the variables are uncorrelated.
- Rank correlation deals with qualitative characteristics.
- Spearman's rank correlation coefficient formula ρ is given by

  $\rho = 1-\frac{6\Sigma d^2}{N(N^2-1)}$ where d = The difference between two ranks = $R_X - R_Y$ and

  N = Number of paired observations.
- Correlation represents linear relationship between the variables but the regression helps to estimate (or predict) one variable by using the other variable.
- Regression lines of

  (i) X on Y is $X-\overline{X} = b_{xy}(Y-\overline{Y})$ (ii) Y on X is $Y-\overline{Y} = b_{yx}(X-\overline{X})$
- The two regression lines passing through their respective means of X and Y
- Calculation of the regression coefficients.

  $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ and $b_{yx} = r\frac{\sigma_y}{\sigma_x}$
- The Properties of regression coefficients.

  (i) $r = \sqrt{b_{yx} \times b_{xy}}$

  (ii) both the regression coefficients cannot be greater than one.

  (iii) Both the regression coefficients have same sign.

09_11th_BM-STAT_Ch-9-EM.indd 233 21-04-2020 12:28:45 PM